KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4127
Image Processing & Computer Vision

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Consider the following two gray scale images (Image *A*, Image *B*) of size $N \times N$. Although they look quite different, their histograms are identical. If each of the images are blurred with the Mask *M*, then (10)
   i) Would the resultant histograms still be the same? Explain.
   ii) Calculate the approximate histograms of the resultant images.

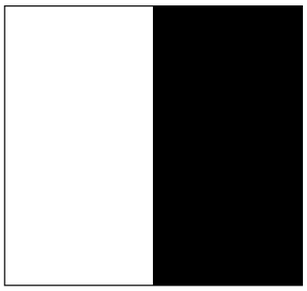


Image: *A*

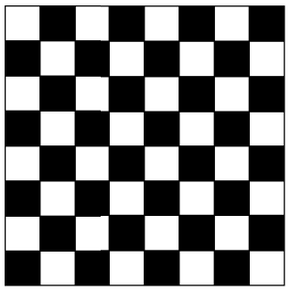


Image: *B*

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

Mask: *M*

b) Let an image *A* has gone through an intensity transformation function $S = T(r) = cr^{\gamma}$, where c, γ are constants to get the output image *B*. Some pixel intensities of *A* and *B* are shown below. (13)
   i) Find the missing input intensities P, Q.

	4		15
		Q	
P		16	

$\longrightarrow T(r) \longrightarrow$

	12		87
		105	
22		96	

Image: *A* Image: *B*

c) Do you think the discrete histogram equalization technique is generally able to produce a flat histogram? Justify your answer. (07)

2. a) Suppose an image with intensities in the range [0,1] has the PDF $P_r(r)$ as shown in Fig. i. Assume continuous quantities and find the transformation in terms of r and z that will transform the intensity levels of the image so that they will have the specified $P_z(z)$ as shown in Fig ii. (12)

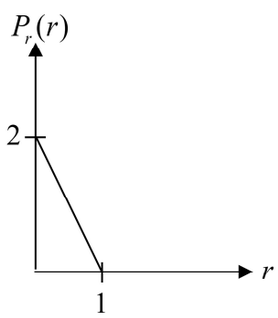


Fig. i

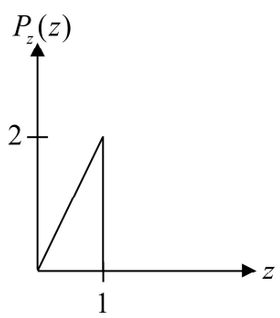


Fig. ii

b) Explain convex deficiency. Is it possible to reduce the hit-or-miss transform to simple erosion? Justify your answer. (12)

c) What does it mean by sampling and quantization in digital image acquisition? (06)

3. a) Consider the following 4×8 image. (15)

21	169	169	243	21	95	132	95
243	169	132	95	109	169	109	132
95	21	132	169	169	95	109	109
95	132	169	21	95	243	132	169

i) Compress the image using Huffman coding.

ii) Calculate the compression ratio and the percentage of the original fixed-length representation that is removed as the coding redundancy.

b) Calculate the morphological thinning on image E using P as the initial structuring element and show the outcome of each step separately. (15)

0	0	0	0	0	0	1	1
1	0	0	0	0	0	1	1
1	1	1	1	1	0	0	0
1	1	1	1	0	1	1	0
1	1	1	1	1	1	0	0
1	1	1	0	1	1	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Image: E

0	0	0
×	1	×
1	1	1

Structuring Element: P

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Let a Gaussian function $G(x,\sigma)=e^{-x^2/2\sigma^2}$ and its derivative G' is sampled at 5 points symmetric to the origin with $\triangle x=1,\sigma=1$. The sampled values of G and G' are used as $1D$ separable kernel to convolve an image in x and y direction, respectively. Find (14)

i) The x and y directional $1D$ kernels; scale and round the values to integer.

ii) The $2D$ kernel from $1D$ kernels.

iii) Note the benefits of using separable $2D$ kernels.

b) Consider the following 4-bit image corrupted with impulse noise. To remove the noise (16)

i) Use a contraharmonic mean filter of size 3×3 with appropriate order Q.

ii) Use a suitable order-statistics filter of size 3×3.

Show results only for the middle row.

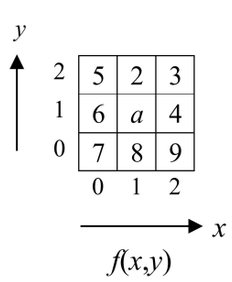
7	6	8	5	4
3	10	9	8	11
8	x	11	12	1
5	8	2	10	7
9	3	4	15	14

Here, $x=\begin{cases}0; & \text{if your roll is an odd number}\\ 15; & \text{Otherwise}\end{cases}$

5. a) Let you are assigned to warp the following image $f(x,y)$ with the transformation matrix T. You are also constrained with the conditions that the transformation should be done using backward mapping technique having bilinear interpolation and the warped image should be large enough to contain all the processed samples. If so (14)

i) Find the size of the warped image.

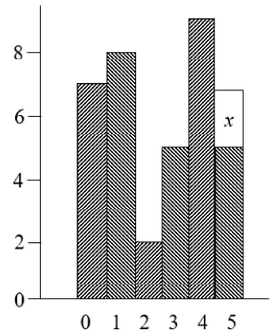
ii) Find the intensity at pixel (1,3) of the warped image.



$$T=\begin{bmatrix}1 & -0.5\\ 0.5 & 1\end{bmatrix}$$

Here, $a=$ last digit of your roll + 1.

b) An image intensity distribution shown by the following bar chart. Find a global optimal threshold for the image using Otsu's method. (16)



Here, $x =$ sum of last two digits of your roll number.
Therefore, $frequency(5) = x + 5$.

6. a) Let you have planned to describe an image texture as an Entropy of Gray Level Co-occurrence Matrix (GLCM). For the given image below (15)
   i) Find the GLCM with displacement $d = [1,0]$.
   ii) Determine the measure of Entropy.

0	1	4	2	2
3	4	3	0	1
1	1	4	2	0
3	2	4	3	1
4	3	0	2	2

b) Using a 2×2 window, estimate the optical flow using Lucas-Kanade method for the following two successive frames of a video sequence. Show result for (1,1) position only. (15)

	0	1	2	3	→ x
0	1	1	0	1	
1	1	9	9	9	
2	1	9	10	9	
3	0	9	9	10	
↓ y					

Frame at t

	0	1	2	3	→ x
0	1	1	0	1	
1	0	a	1	0	
2	1	1	9	9	
3	0	1	9	9	
↓ y					

Frame at $t+1$

Here, $a = \begin{cases} 0; & \text{if your roll is even} \\ 1; & \text{Otherwise} \end{cases}$